Project Synopsis: Explainable and Uncertainty-Aware Deep Learning for Diabetic Retinopathy Grading with Retinal Image Quality Control

Abstract

This project proposes a research prototype that analyzes retinal fundus photographs for diabetic-retinopathy screening support. The system will first assess whether an image is suitable for automated analysis, then estimate a retinopathy severity grade for images that pass quality checks. It will also report prediction uncertainty and provide a visual explanation aid. Rather than forcing a prediction on every image, the pipeline will be able to return "unable to assess" when image quality is poor or model confidence is insufficient.

The prototype will use a suitably licensed, publicly available retinal-image dataset. Image preprocessing, data splitting, model training, and evaluation will be performed in Python using a deep-learning framework. A web interface will demonstrate image submission, quality assessment, grading, and result presentation. Evaluation will compare a baseline classifier with a transfer-learning model, using class-sensitive performance measures, calibration, and tests under selected image-quality perturbations. The system is an academic prototype and is not intended for diagnosis or clinical decision-making.

1. Introduction

Automated analysis of retinal photographs is a useful computer-vision problem with a direct healthcare context. A simple image classifier, however, can conceal important failure modes: an image may be blurred or poorly illuminated, the model may be uncertain, and a single overall accuracy score may hide weak performance on less common severity grades. This project therefore treats image quality control and uncertainty handling as part of the system rather than as optional additions to classification.

The proposed work will develop an end-to-end prototype for diabetic-retinopathy severity grading from fundus images. It will communicate its output as a screening-support estimate, expose relevant limitations, and allow an "unable to assess" outcome instead of presenting an unreliable prediction as definitive.

2. Problem Statement

Many educational image-classification demonstrations produce a disease label without checking whether the input image is analyzable, showing how uncertain the output is, or evaluating whether errors differ across severity classes. The project will address this engineering and evaluation problem by designing a pipeline that combines retinal image quality assessment, ordered severity classification, uncertainty-aware abstention, and a simple user-facing interface.

3. Aim

To design, implement, and evaluate an explainable, uncertainty-aware deep-learning prototype for diabetic-retinopathy grading from retinal fundus photographs, incorporating image-quality control and an option to abstain from grading.

4. Objectives

1. Identify and document a public retinal fundus image dataset whose access and usage terms permit the proposed academic work.
2. Define the target grading labels and inspect class balance, image characteristics, and available patient-level identifiers.
3. Develop a reproducible preprocessing pipeline and data split that prevents images from the same patient appearing across training and test sets when patient identifiers are available.
4. Implement an image-quality assessment stage that can identify images that should be rejected or referred for human review.
5. Train a transparent baseline model and at least one transfer-learning or convolutional neural-network model for severity grading.

6. Add an uncertainty-based abstention rule so the system can return "unable to assess" rather than a low-confidence grade.
7. Provide a visual explanation aid and clearly communicate that it is an aid to model interpretation, not proof of clinical reasoning.
8. Evaluate classification performance, image-quality rejection, calibration, and robustness using a held-out test set and defined metrics.
9. Build a prototype interface for uploading an image and viewing the quality result, predicted grade or abstention, uncertainty information, and limitations.
10. Document ethical, privacy, data-use, and clinical-validation limitations, and provide instructions to reproduce the experiments.

5. Research Questions

1. How does the selected transfer-learning model perform compared with a simple baseline for retinal-image severity grading on a held-out test set?
2. Does adding image-quality screening identify unsuitable images before grading, and what trade-off does it create between coverage and classification performance?
3. How do confidence calibration and an abstention threshold affect the number of images receiving a prediction and the errors among those predictions?
4. How sensitive is model performance to selected image-quality perturbations, such as blur, brightness changes, or reduced resolution?

6. Scope and Assumptions

Included

- Retinal fundus photographs from one permitted public dataset.
- A defined set of diabetic-retinopathy severity grades available in that dataset.
- Image preprocessing, quality assessment, supervised model training, evaluation, and a demonstration interface.
- A retrospective experimental evaluation on a held-out dataset split.

Excluded

- Diagnosis or treatment recommendations for individual patients.
- Clinical deployment, prospective clinical validation, or claims of clinical benefit.
- Integration with hospital records or identifiable patient data.
- Generalization claims across cameras, hospitals, or populations not represented in the available data.
- Autonomous referral or treatment decisions.

The implementation assumes that the dataset labels and terms of use are adequate for a student research prototype. If the available dataset lacks patient identifiers, the project will document that limitation and use the strongest feasible leakage-control strategy based on the provided metadata.

7. Proposed Methodology

7.1 Dataset selection and audit

Select a publicly available retinal-image dataset after reviewing its documentation, permitted uses, label definitions, and metadata. Record the number of usable images, label distribution, image dimensions, and any available patient or acquisition identifiers. Exclude no image silently: define preprocessing and exclusion rules in advance and report how many images are affected.

7.2 Data preparation

Resize images to the model input dimensions and apply a documented normalization procedure. Training-time augmentation may be used to reduce sensitivity to benign variation, but transformations must preserve the meaning of

the clinical labels. Separate training, validation, and test data before fitting preprocessing choices or model parameters. Where identifiers permit, split by patient rather than by image to reduce leakage.

7.3 Image-quality assessment

Implement a quality-control stage that identifies images unsuitable for grading. Depending on dataset labels and project time, this can use either documented image-quality labels or a defined image-quality classifier/heuristic pipeline. If no reliable quality labels are available, the project will treat quality checks as a limited engineering component and validate them using a small, manually reviewed sample rather than presenting unvalidated quality scores as ground truth.

7.4 Severity classification

Train a simple baseline, such as logistic regression on image features or a compact CNN, followed by a transfer-learning model. Use the same patient-aware data partitions and evaluation protocol for comparisons. Since severity grades are ordered, report the full confusion matrix and consider an ordinal-sensitive error measure in addition to conventional classification metrics.

7.5 Uncertainty and abstention

Use a validation set to select a confidence or uncertainty threshold. At inference, return a grade only when the image passes the quality gate and the prediction meets the chosen threshold; otherwise return "unable to assess" or "review required." Report both the fraction of images receiving predictions (coverage) and performance among those predictions, so improved accuracy is not presented without its associated abstention rate.

7.6 Explainability

Provide a visual explanation method, such as a saliency or activation map, for demonstration and qualitative inspection. Present it as an approximate indication of image regions associated with model output, not as evidence that the model has identified a clinically valid lesion or reasoned like a clinician. Inspect examples of plausible and implausible maps and include representative failure cases.

7.7 Evaluation

Evaluate only on the held-out test partition after model and threshold selection are complete. Report:

- Macro-F1 and per-grade precision, recall, and F1.
- Confusion matrix and overall accuracy as a secondary summary.
- Sensitivity/recall for each grade, with particular attention to missed higher-severity cases.
- Calibration using a reliability plot and a suitable calibration summary.
- Image-quality screening performance if appropriate labels or a reviewed sample are available.
- Coverage and error rates at the selected abstention threshold.
- Robustness under predefined blur, brightness, and resolution perturbations, compared with unmodified images.
- Inference time and basic application functionality for the prototype.

Results will be reported with dataset limitations and without interpreting model metrics as evidence of clinical effectiveness.

8. System Architecture

1. **User interface**: accepts an image and presents results and limitations.
2. **Input validation**: checks file type, dimensions, and whether the image can be processed.
3. **Preprocessing service**: applies the same documented transformations used during training.
4. **Quality-control module**: accepts or rejects the image for grading.
5. **Prediction module**: returns a severity estimate and uncertainty information, or abstains.
6. **Explanation module**: creates a visual aid for the displayed prediction.

7. **Results layer**: displays the grade, quality status, uncertainty, explanation, and non-diagnostic notice.

Uploaded images should not be retained by default. If temporary storage is needed during inference, the prototype should delete files after processing and avoid collecting personal identifiers.

9. Technology Stack

- **Programming language**: Python
- **Data processing**: pandas, NumPy, OpenCV, Pillow
- **Machine learning**: PyTorch or TensorFlow/Keras; scikit-learn for baselines and metrics
- **Explainability**: a compatible saliency or activation-map method
- **API**: FastAPI, if the model is separated from the interface
- **Prototype interface**: Streamlit for a fast academic demo, or React for a separate frontend
- **Experiment tracking**: versioned configuration files and saved model checkpoints; optional lightweight experiment tracking
- **Source control**: Git

The project can begin on CPU with a compact model and reduced image resolution; GPU use is optional and depends on available hardware.

10. Expected Deliverables

1. Dataset and label audit with documented data-use conditions.
2. Reproducible preprocessing and patient-aware split, where identifiers permit.
3. Baseline and deep-learning model implementations.
4. Image-quality and uncertainty/abstention components.
5. Evaluation report with defined metrics, robustness tests, calibration, and failure analysis.
6. Working prototype interface and demonstration.
7. Source code, setup instructions, and experiment configuration.
8. Final dissertation/report describing methods, results, limitations, and ethical boundaries.

11. Proposed Work Plan

Stage	Work	Indicative duration
1	Confirm scope, review dataset documentation, finalize labels and protocol	1–2 weeks
2	Audit data, create preprocessing and leakage-aware splits	1–2 weeks
3	Train baseline and establish initial evaluation	2 weeks
4	Train and tune the deep-learning model using validation data	2–3 weeks
5	Implement quality control, uncertainty, and abstention	2 weeks
6	Perform final evaluation, robustness checks, and failure analysis	2 weeks
7	Build or complete the demonstration interface	1–2 weeks
8	Write report, prepare demo, and incorporate supervisor feedback	2 weeks

The schedule is indicative and should be adapted to the institution's project calendar and the size and format of the chosen dataset.

12. Risks and Mitigations

- **Dataset access or terms are unsuitable**: verify conditions before implementation and select an alternative dataset or narrow the scope.
- **Class imbalance**: report per-class results, use appropriate training strategies, and avoid relying on overall accuracy.
- **Data leakage**: use patient-level splitting when identifiers are available; clearly document limitations when they are not.
- **Unreliable quality labels**: treat quality assessment as exploratory and validate against a manually reviewed subset if feasible.
- **Limited compute**: use a compact pretrained model, smaller input size, and staged experiments.
- **Overstated explainability**: describe visual maps as aids, inspect failure cases, and avoid causal or clinical interpretations.
- **Overclaiming generalization**: restrict conclusions to the dataset and evaluation setting; do not claim clinical readiness.

13. Ethical and Safety Considerations

The work will use data only under the dataset's stated terms and institutional requirements. The prototype will not provide a diagnosis, treatment recommendation, or autonomous clinical decision. Its interface will state that outputs are experimental and require qualified human interpretation. Privacy will be supported by minimizing data collection and avoiding image retention by default. Any evaluation involving human participants or expert image review will follow applicable institutional approval and consent requirements.

14. Conclusion

This project combines medical-image classification with image-quality control, uncertainty-aware abstention, calibration, and robustness evaluation. Its contribution is not a claim that diabetic-retinopathy classification is a new task; it is a carefully scoped, reproducible prototype that examines when a model should provide a result and when it should decline to do so. The planned deliverables provide both a working software system and a measurable machine-learning evaluation, while keeping the work within a non-clinical academic boundary.